EXAM PAPERS PRACTICE

Q1.

The random variable X has an exponential distribution with mean μ. The cumulative distribution function of X for $x \geq 0$ is given by

$$F(x) = 1 - e^{-\frac{1}{\mu}x}$$

(a) Find an exact expression for the interquartile range of X in terms of μ.

(5)

(b) Prove, by integration, that $E(X^2) = 2\mu^2$.

(4)

(c) Show that the standard deviation of X is less than the interquartile range of X.

(3)

(Total 12 marks)

Q2.

A random variable X has an exponential distribution with probability density function $f(x)$, where

$$f(x) = \begin{cases} ke^{-kx} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

and k is a constant.

(a) Given that $E(X) = \frac{1}{k}$, find:

(i) using integration, $E(X^2)$;

(ii) $\mathrm{Var}(X)$.

(6)

(b) (i) Derive the cumulative distribution function, $F(x)$, of X for $x \geq 0$.

(3)

(ii) Hence find, in terms of k, the **exact** value of the 90th percentile of X.

(3)

(c) A machine has two essential components, the lifetimes of which follow exponential distributions with means a hours and $3a$ hours. The machine will stop if either component fails. The failures of the two components may be taken to be independent.

Find the probability that the machine continues to work for at least a hours from the start, giving your answer in the form e^q, where q is a rational number to be determined.

(4)

(Total 16 marks)

Q3.

(a) The continuous random variable X follows an exponential distribution if it has a probability density function

$$f(x) = \begin{cases} ke^{-kx} & x \geq 0 \\ 0 & \textit{otherwise} \end{cases}$$

where k is a positive constant.

(i) Prove that the mean value, μ, of X is $\frac{1}{k}$.

(3)

(ii) Find, in terms of k, the median value, m, of X and hence show that $m < \mu$.

(5)

(b) The number of radioactive particles striking a screen in a time period of length t seconds follows a Poisson distribution with mean $\frac{t}{\lambda}$, where λ is a constant.

(i) Write down the probability that no particles strike the screen in a period of t seconds.

(1)

(ii) The random variable T is defined as the length of time, in seconds, between successive radioactive particles striking the screen.

(**A**) Show that

$$P(T < t) = 1 - e^{-\frac{t}{\lambda}}$$

(2)

(**B**) Hence, by finding the probability density function of T, state the distribution of T.

(2)

(Total 13 marks)

Q4.

The continuous random variable X is modelled by an exponential distribution with probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

(a) Write down the cumulative distribution function, F(x).

(2)

(b) Show that the **exact** value of the interquartile range of X is given by $\frac{1}{\lambda}$ ln3.

(5)

(c) (i) Use integration to prove that $E(X^2) = \frac{2}{\lambda^2}$.

(4)

(ii) Hence, given that $E(X) = \frac{1}{\lambda}$, show that $Var(X) = \frac{1}{\lambda^2}$.

(1)

(d) (i) Find the **exact** value of λ for which the value of the interquartile range of X is four times the value of the variance of X.

(2)

(ii) Describe what happens to the interquartile range of X as λ increases without bound.

(1)
(Total 15 marks)

Q5.

(a) A continuous random variable X has probability density function

$$f(x) = \lambda e^{-\lambda x} \text{ for } x \geq 0$$

Prove that the mean value of X is $\frac{1}{\lambda}$.

(4)

(b) The lifetime of a component in a machine is T hours, where T has probability density function

$$f(t) = \frac{1}{a} e^{-\frac{t}{a}} \text{ for } t \geq 0$$

The mean lifetime of these components is known to be 62.5 hours.

(i) Find the value of $\frac{1}{a}$.

(2)

(ii) Calculate the probability that a component will last for at least 80 hours.

(4)

(iii) Given that a component has lasted for 80 hours, find the probability that it will last for a further 20 hours.

(3)
(Total 13 marks)

Q6.

The lifetime, T hours, of a certain electrical component is exponentially distributed with mean 800.

(a) (i) State, in full, the probability density function of T.

(2)

(ii) Write down the numerical value of the standard deviation of T.

(1)

(iii) Calculate the probability that one of these components, chosen at random, is still working after 1000 hours.

(3)

(iv) Given that one of these components is still working after 1000 hours, calculate the probability that it is still working after a further 1000 hours.

(3)

(b) A piece of equipment contains four of these components. The design is such that it is still functional when at least two of the four components are working. Calculate the probability that the piece of equipment is still functional after 1000 hours.

(4)
(Total 13 marks)

Q7.

Between 8 am and 6 pm, the time, T minutes, between successive arrivals of vehicles for fuel at a village garage may be assumed to have an exponential distribution with a mean of 8.

(a) Write down the numerical value for the standard deviation of T.

(1)

(b) Calculate the probability that the time between successive arrivals of vehicles for fuel at the garage is between 5 minutes and 15 minutes.

(3)

(c) Between 6 pm and 9 pm, the time, S minutes, between successive arrivals of vehicles for fuel at the garage may be assumed to be independent of T and to have an exponential distribution with a mean of 15.

Calculate the probability that no vehicles arrive for fuel at the garage between 5.45 pm and 6.15 pm.

(5)
(Total 9 marks)